Numbers

2, 5, −10,
−0.25, $\frac{11}{7}$,
8.5, $\frac{-22}{4}$

Variables

In case of uncertainity, Letters are assumed to be numbers.

e.g. → x, y, z, a, b etc.

Expressions or Polynomials

{Variables + Numbers}

e.g. → $3p+4$,
$3y-11$,
$8p+2q-4$, etc.

Equation →

When we use "=" (equal sign) in an expression or polynomial, it is called an Equation.

Let's take an example,

- Let the age of Aman → P
- Age of Firoz is twice the age of Aman → 2P
- Age of Sushila is 7 more than age of Aman → P+7
- Age of Mother is 11 less than 4 times the age of Aman. → 4P − 11

Here, (P), (2P), (P+7), (4P−11) these are all expressions / Polynomials.

- Age of Aman is 22 less than Age of Firoz

$\boxed{(P) = 2P - 22}$ → Now, this is an Equation.

Degree of an Equation →

Maximum power of a variable in an equation is known as degree of an equation.

Degree 1 → Linear Equation

Degree 2 → Quadratic Equation

Degree 3 → Cubic Equation

Example →

- $3x + 4y = 7$ Maximum Power 1 → Degree 1 { Linear Equation }

- $4x + 3x^2 = 9$ Maximum Power 2 → Degree 2 { Quadratic Equation }

- $9y^2 + 2x^3 = 0$ Maximum Power 3 → Degree 3 { Cubic Equation }

- $4x^3 + 2x^2 + 5x + 7 = 0$ Maximum Power 3 → Degree 3 { Cubic Equation }

$$\boxed{\text{Number of Zeroes} = \text{Number of Roots} = \text{Number of Solutions}}$$

This means, the number of values which satisfies the equation, when we put the value at the place of variable.

These are the roots or zeroes of equation.

- Degree of an Equation decides the number of roots.

⇒ Linear Equation has 1 root.

⇒ Quadratic Equation has 2 roots.

⇒ Cubic Equation has 3 roots.

Linear Equation

Linear Equation in two variables.

$$2x - 3y + 10 = 0$$

- If $x = 1$, $2(1) - 3y + 10 = 0$

$$12 = 3y$$
$$\underline{y = 4}$$

- If $x = 2$, $2(2) - 3y + 10 = 0$

$$14 = 3y$$
$$y = \frac{14}{3}$$

- If $x = 4$, $2(4) - 3y + 10 = 0$

$$18 = 3y$$
$$\underline{y = 6}$$

In single linear equation of two variables, there can be infinite number of solutions.

So, to solve (n) variables in linear equations, we need (n) independent linear equations.

- In case of 1 variable, 1 linear equation is required
- To solve two variables, 2 linear equations (independen[t]) are required.
- To solve three variables (i.e., x, y, z), 3 independent linear equations are required.

Quadratic Equation

A quadratic equation in the variable x is an equation of the form $ax^2+bx+c=0$, where a, b, c are real numbers, $a \neq 0$.

Roots / Zeroes of the equation →

The values of x satisfying the quadratic equation $ax^2+bx+c=0$ are called solutions, zeroes or roots of the quadratic equation.

★ Quadratic Expression

$$ax^2+bx+c=0$$

{ a, b, c are coefficients }

- c is the constant term.
- $\underline{x^2}$: a $\quad$ $\underline{x^1}$: b $\quad$ $\underline{x^0}$: c

★ Quadratic Equation

$$ax^2+bx+c=y$$

$$ax^2+bx+c=0$$

when the value of y becomes zero, we find roots.

→ Solutions of Quadratic Equation

$$x^2-x-6=0$$

$$x^2-3x+2x-6=0$$

$$x(x-3)+2(x-3)=0$$

$$(x-3)(x+2)=0 \longrightarrow \quad x-3=0, \quad x+2=0$$

$$\left.\begin{array}{l} x=+3 \\ x=-2 \end{array}\right\}$$

+3 and −2 are the roots of the Quadratic Equation.

- For a Quadratic Equation,

$$ax^2 + bx + c = 0$$

Sum of roots $\rightarrow -(b/a)$

Product of roots $\rightarrow +(c/a)$

$\left\{ \begin{array}{l} a, b \text{ and } c \\ \text{are the} \\ \text{coefficients} \end{array} \right\}$

Examples $\rightarrow$

- $x^2 - 5x + 6 = 0$

$\underline{a=1} \quad \underline{b=-5} \quad \underline{c=6}$

Sum of Roots $\rightarrow -\left(\frac{-5}{1}\right) \Rightarrow +5 \quad \left(-\frac{b}{a}\right)$

Product of Roots $\rightarrow \frac{6}{1} \Rightarrow +6 \quad \left(\frac{c}{a}\right)$

- $x^2 - 14x + 40 = 0$

$\underline{a=1} \quad \underline{b=-14} \quad \underline{c=40}$

Sum of Roots $\rightarrow -\left(\frac{-14}{1}\right) \rightarrow +14 \quad \left(-\frac{b}{a}\right)$

Product of Roots $\rightarrow \left(\frac{6}{1}\right) \rightarrow +6 \quad \left(\frac{c}{a}\right)$

Equation	Sum of Roots $\rightarrow -\frac{b}{a}$	Product of Roots $\rightarrow +\frac{c}{a}$
$x^2+3x-10=0$	$-\left(\frac{+3}{1}\right) \Rightarrow -3$	$\frac{-10}{1} \Rightarrow -10$
$x^2-11x+24=0$	$-\frac{b}{a} \Rightarrow -\left(\frac{-11}{1}\right) \Rightarrow +11$	$\frac{c}{a} = \frac{+24}{1} = +24$
$2x^2-16x-256=0$	$-\left(\frac{-16}{2}\right) = +8$	$\frac{-256}{2} = -128$
$3x^2+12x-96=0$	$-\left(\frac{12}{3}\right) = -4$	$\frac{-96}{3} = -32$
$x^2-9x+18=0$	$-\left(\frac{-9}{1}\right) = +9$	$\frac{18}{1} = +18$
$5x^2-35x-90=0$	$-\left(\frac{-35}{5}\right) = +7$	$\frac{-90}{5} = -18$
$x^2+10x-600=0$	$-\left(\frac{10}{1}\right) = -10$	$\frac{-600}{1} = -600$
$7x^2-49x+70=0$	$-\left(\frac{-49}{7}\right) = +7$	$\frac{+70}{7} = +10$

Equation	Sum of Roots $= \frac{-b}{a}$	Product of Roots $= \frac{c}{a}$
$2x^2 - 18x + 36 = 0$	$-\left(\frac{-18}{2}\right) = +9$	$+\frac{36}{2} = +18$
$6x^2 - 30x - 84 = 0$	$-\left(\frac{-30}{6}\right) = +5$	$-\frac{84}{6} = -14$
$x^2 - 8x + 15 = 0$	$-\left(\frac{-8}{1}\right) = +8$	$+15/1 = +15$
$2x^2 + 14x - 36 = 0$	$-\left(\frac{+14}{2}\right) = -7$	$-\frac{36}{2} = -18$
$2x^2 - 2x - 60 = 0$	$-\left(\frac{-2}{2}\right) = +1$	$-60/2 = -30$
$4x^2 - 22x + 30 = 0$	$-\left(\frac{-22}{4}\right) = +\frac{11}{2}$	$+\frac{30}{4} = +\frac{15}{2}$
$5x^2 - 125x + 780 = 0$	$-\left(\frac{-125}{5}\right) = +25$	$+780/5 = +156$
$x^2 + 53x + 660 = 0$	$-\left(\frac{53}{1}\right) = -53$	$+660/1 = +660$
$x^2 + 2x - 255 = 0$	$-\left(\frac{2}{1}\right) = -2$	$-\frac{255}{1} = -255$
$2x^2 + 34x + 144 = 0$	$-\left(\frac{34}{2}\right) = -17$	$+144/2 = +72$
$x^2 + 31x - 360 = 0$	$-\left(\frac{31}{1}\right) = -31$	$-\frac{360}{1} = -360$
$3x^2 - 81x + 216 = 0$	$-\left(-81/3\right) = +27$	$+216/3 = +72$
$x^2 - 5x + 6 = 0$	$-\left(-5/1\right) = +5$	$+6/1 = +6$
$x^2 - 7x + 12 = 0$	$-\left(-7/1\right) = +7$	$+12/1 = +12$
$x^2 - 14x + 40 = 0$	$-\left(-14/1\right) = +14$	$+40/1 = +40$
$x^2 + 11x + 24 = 0$	$-\left(11/1\right) = -11$	$+24/1 = +24$
$x^2 + 9x + 14 = 0$	$-\left(9/1\right) = -9$	$+14/1 = +14$
$3x^2 - 5x - 2 = 0$	$-\left(-5/3\right) = +5/3$	$-\frac{2}{3} = -2/3$
$x^2 + 3x - 10 = 0$	$-\left(3/1\right) = -3$	$-10/1 = -10/1$
$x^2 + 5x + 6 = 0$	$-\left(5/1\right) = -5$	$+6/1 = +6$

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$x = \frac{-b + \sqrt{b^2-4ac}}{2a}, \quad \frac{-b-\sqrt{b^2-4ac}}{2a}$$

where, $b^2 - 4ac$ = Discriminant

Cubic Equation

In algebra, a cubic equation in one variable is an equation of the form:

$$\boxed{ax^3 + bx^2 + cx + d = 0}, \text{ where } a \neq 0.$$

Let roots of this equation be m, n, p.

Sum of Roots $(m + n) = -b/a$

Sum of Product of Pair of roots $(mn + np + mp) = c/a$

Product of Roots $(m.n.p) = -d/a$

Equation	Sum of Roots $(m+n+p)$	Product of Roots $(m.n.p)$	Sum of Product of Pair of Roots $(mn+np+mp)$
• $x^3-7x^2+14x-8=0$	$+7$ $\left(-b/a\right)$	$+8$ $\left(-d/a\right)$	$+14$ $\left(c/a\right)$
• $x^3-15x^2+66x-80=0$	$+15$	$+80$	$+66$
• $x^3+6x^2-25x+18=0$	-6	-18	-25
• $2x^3+18x^2+4x-96=0$	-9	$+48$	$+2$

Solving Quadratic Equations →

$\{ax^2+bx+c=0\}$

1) $x^2-3x+2=0$

Step 1 → (Coefficient of x^2) x (constant)

Step 2 → Find two numbers, whose
Multiplication → Step 1
Addition → coefficient of x

Step 3 → Change sign of both numbers of Step 2

Step 4 → Divide both numbers (Step 3) with coefficient of x^2.

$x^2-3x+2=0$

1) $(1)\times(2) = 2$

2) Multiply = $+2$
Addition = -3
$(-2,-1)$

{Those numbers are -2 and -1}

3) $+2, +1$

4) $+\frac{2}{1}, +\frac{1}{1}$ ⟹ Roots of this Quadratic Equation are $(+2, +1)$

$\boxed{x^2-5x+6=0}$

1) $(1)\times(6) = +6$

2) Multiply = $+6$
Addition = -5
$(-3,-2)$

3) $+3, +2$

4) $\left(+\frac{3}{1}, +\frac{2}{1}\right) \Rightarrow (+3, +2)$

$\boxed{2x^2-11x+15=0}$

1) $(2)\times(15) = 30$

2) Multiply = (30)
Addition = (-11)
$(-6,-5)$

3) $+6, +5$

4) $\left(+\frac{6}{2}, +\frac{5}{2}\right) \Rightarrow (+3, +2.5)$

$\boxed{x^2+7x+12=0}$

1) $(1)\times(12) = 12$

2) Multiply $= +12$

Addition $= +7$

$(+3, +4)$

3) $(-3, -4)$

4) $\left(\frac{-3}{1}, -\frac{4}{1}\right) \Rightarrow (-3, -4)$

Roots

$\boxed{x^2+9x+14=0}$

1) $(1)\times(14) = 14$

2) Multiply $= +14$

Addition $= +9$

$(+7, +2)$

3) $-7, -2$

4) $-\frac{7}{1}, -\frac{2}{1} \Rightarrow (-7, -2)$

Roots

$\boxed{x^2-7x-18=0}$

1) $(1)\times(-18) = -18$

2) Multiply $= -18$

Subtract $= -7$

$(-9, +2)$

3) $(+9, -2)$

4) $+\frac{9}{1}, -\frac{2}{1} \Rightarrow (9, -2)$

Roots

$\boxed{3x^2-5x-2=0}$

1) $(3)\times(-2) = -6$

2) Multiply $= -6$

Subtract $= -5$

$(-6, +1)$

3) $6, -1$

4) $\frac{6}{3}, \frac{-1}{3} \Rightarrow \left(2, -\frac{1}{3}\right)$

Roots

$\boxed{x^2+3x-10=0}$

1) $(1)\times(-10) = -10$

2) Multiply $= -10$

Subtract $= +3$

$(+5, -2)$

3) $-5, +2$

4) $-\frac{5}{1}, +\frac{2}{1} \Rightarrow (-5, 2)$

Roots

$\boxed{2x^2+7x-4=0}$

1) $(2)\times(-4) = -8$

2) Multiply $= -8$

Subtract $= +7$

$(+8, -1)$

3) $-8, +1$

4) $-\frac{8}{2}, +\frac{1}{2} \Rightarrow \left(-4, \frac{1}{2}\right)$

Roots